

TECHNICAL WHITE PAPER

Building a Deterministic Six-Degree-of-Freedom
Flight Simulation Model with Atmospheric Interaction
on Constrained Hardware

Table-Driven Aerodynamics with Static and Dynamic Stability Derivatives,
Shared Atmosphere and Wind Service, Air-Breathing Engines with Installation Moments,
Variable-Geometry Wing Sweep, Control Mixer, and a Flyable Physical Airframe
under AUGMENTED and AUG_OFF Flight-Control Modes
for the F-65 Megawing Simulation Engine on the MEGA65

Audience:	Game designers and software engineers (C / 45GS02 assembly)
Platform:	MEGA65 — 45GS02 + VIC-IV; fixed-point arithmetic only
Simulation rate:	Exactly 100 Hz deterministic tick
Units:	Imperial; numeric registry is authoritative for every scalar
Version:	3.3 — FROZEN governing engineering model for Phase 2
Date:	23 August 2026
Project authority:	Revision 1.5.1 Architecture Invariants; Gameplay Supplement Draft 0.2
Status:	FROZEN — Phase 1 consumes interface shapes; Phase 2 implements this model

This paper is the frozen governing engineering model for the F-65 flight-simulation physics that Phase 2 will implement. The F-65 is a physically flyable airplane. Default mode is modern F/A-18-like augmentation. AUG_OFF removes G-command, roll-rate command, AoA/G protection, and artificial FCS damping but retains trim, hydraulics, actuator limits, non-augmenting control mixing/gearing, and natural aerodynamic static and dynamic stability. ADLC is an approach overlay that disengages without state snap. Phase 1 must expose the interface shapes, state ownership, table-residency, and oracle contracts defined here so that Phase 2 plugs in without redesign. Phase 1 uses bounded fixtures and proxies; actual coefficient tables and schedules remain Phase-2-gated. Architecture and Gameplay must propagate FCSSMode / FCSSStatus under configuration control before Phase 1 freezes public interfaces.

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1. Purpose, Project Context, and Phase Relationship

The F-65 Megawing programme requires a consequential table-driven 6DOF rigid-body model and matching 3DOF weapon models that run deterministically at 100 Hz on the MEGA65. The physical airframe must remain controllable with control augmentation OFF. This white paper is the frozen engineering model for flight dynamics, atmosphere, engines, FCS behaviour, and validation.

R0-D/E/F → Measured-Limits Revision → PHASE 1 (integrated engine harness: scheduler, pools, memory, input, audio, renderer, resources, snapshots, replay, faults, plus proxy/model interfaces) → PASS → PHASE 2 (actual F-65 6DOF / aero / FCS / engines using this paper as the governing engineering model).

Phase 1 consumes record shapes, state ownership, 100 Hz assumptions, table residency, control-surface pipeline, FCSMode/FCSStatus records, oracle interfaces, and budget expectations from this document. Phase 1 uses bounded fixtures and proxies; it does not author production coefficient tables or envelope numbers. Architecture and Gameplay currently still contain Assisted/Manual rate-command language. That language is superseded for engineering purposes by the FCSMode / FCSStatus model defined here. A configuration-controlled revision must propagate these contracts before Phase 1 freezes the public interface definitions that Phase 2 will consume. Phase 1 interface generation follows this paper's record shapes, not the older Assisted/Manual field names.

2. Constrained-Hardware Reality

45GS02, exact 100 Hz, no floating-point in authoritative paths, no general heap, fixed pools. Hot coefficient sets for live aircraft types have a deterministic mission-load residency plan. C primary; assembly evidence-driven.

3. Frames, Units, Mass, and State

NED world; body +X forward / +Y right / +Z down. Numeric registry is sole authority for every scalar. Weight is force (lbf); mass is weight/g (slug) when used in $F = ma$. Core state includes body velocity, quaternion (+ cached DCM), rates p/q/r, mass, configuration/damage, actual left and right wing-sweep positions, and actual individual control-surface positions. Aerodynamics consume actual surfaces, never raw pilot command.

4. Atmosphere, Winds, and Airspeed Definitions

Shared deterministic service: T, P, ρ , a, g, and wind from surface / middle / high-altitude layers with fixed-point altitude interpolation. $V_{air} = V_{ground} - W(h)$. $TAS = |V_{air}|$; $Mach = TAS/a$; $q = \frac{1}{2} \rho TAS^2$; EAS from q ; KIAS/CAS from the approved pressure conversion (primary pilot-facing handling and autothrottle speed). Same KIAS $\approx$ same aerodynamic loading; same TAS produces less q as density falls.

5. Air-Breathing Engines — Force and Moment

Each engine independently schedules thrust vs altitude/ σ , Mach, throttle, AB. Installation position is aircraft data: $F = \sum F_i$, $M = \sum (r_i \times F_i)$. Engine-out produces real yaw without special-case code.

6. Six-Degree-of-Freedom Rigid-Body Dynamics

6.1 Translation (body axes)

$$\begin{aligned} \dot{u} &= r v - q w + F_x / m \\ \dot{v} &= p w - r u + F_y / m \\ \dot{w} &= q u - p v + F_z / m \end{aligned}$$

6.2 Rotation and scheduled inertia

Inertia from a sweep-bin $\times$ mass-bin schedule (I_{xx} , I_{yy} , I_{zz} , I_{xz}) using the tensor convention of Appendix A.2. The target treats scheduled inertia quasi-statically; the explicit dI/dt contribution is omitted unless the high-precision oracle shows a meaningful trajectory error.

6.3 α , β , nondimensional rates

$$\begin{aligned} \alpha &= \text{atan2}(w_a, u_a), \quad \beta = \text{atan2}(v_a, \sqrt{u_a^2 + w_a^2}), \quad V = |V_{air}| \\ p &= p b / (2V), \quad q = q c / (2V), \quad r = r b / (2V), \quad q = \frac{1}{2} \rho V^2 \end{aligned}$$

7. Table-Driven Aerodynamics — Static and Dynamic Stability

Additive organisation: $C = C_{base}(\alpha, \beta, Mach, sweep, config) + C_{\delta} + C_{rate}(p, q, r) + C_{damage} + C_{asym_sweep} + \dots$ Dynamic derivatives mandatory for AUG_OFF: C_{m_q} , C_{l_p} , C_{n_r} , C_{l_r} , C_{n_p} . Natural static stability mandatory: Phase-2 data shall establish adequate longitudinal, directional, and lateral/dihedral static stability throughout F65_UNAUGMENTED_FLIGHT_ENVELOPE, independent of FCS augmentation. Stall/separation uses a bounded deterministic state representation; thresholds remain Phase-2 TBD.

8. Flight-Control Architecture

The physical airplane is flyable with augmentation OFF. Pilot-selected mode, degradation capability, approach control, input context, and autothrottle are separate concerns.

8.1 Separated concepts

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ControlContext = Deck / TFL / Normal Flight / Combat
FCSMode (commanded / pilot-selected) = AUGMENTED / AUG_OFF
FCSStatus / Capability = NORMAL / DEGRADED / DIRECT_ONLY
+ per-axis and per-sensor capability flags as required by the damage model
ADLCState = OFF / CAPTURE / TRACK / DEGRADED
AutothrottleState = independent
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FCSMode records the commanded / pilot-selected control mode: AUGMENTED or AUG_OFF. FCSStatus records available augmentation capability: NORMAL, DEGRADED, or DIRECT_ONLY. Effective control behaviour is derived deterministically from both. DIRECT_ONLY status forces direct behaviour regardless of a commanded AUGMENTED mode, without rewriting the commanded FCSMode. A failure therefore does not conceptually "select" the pilot's mode; the cockpit switch may still indicate AUGMENTED while the system announces DIRECT ONLY. ControlContext governs HOTAS semantics and presentation only — Normal Flight is a context, not an FCS law.

8.2 Command hierarchy with control mixer

InputCommandFrame → Control Context → FCS Mode Manager (commanded mode + capability status) → optional ADLC longitudinal/throttle overlay → command generation → non-augmenting Control Mixer / approved control gearing → Hydraulic authority → Actuator dynamics → actual individual ControlSurfaceState → 6DOF aerodynamics.

The control mixer remains present under AUG_OFF. It may implement differential stabilator, aileron–stabilator mixing, rudder-authority scheduling, and q ■/config/sweep gearing. It contains no closed-loop stability augmentation, no G-command feedback, and no artificial damping.

8.3 AUGMENTED mode

Modern F/A-18-like character. Longitudinal stick → scheduled normal acceleration with pitch-rate/AoA feedback where appropriate. Lateral stick → scheduled roll rate. Yaw damping and coordination active. Automatic trim as designed. G/AoA limiting and authority scheduling by configuration, sweep, q ■, Mach, hydraulics, and damage. No altitude hold; no artificial lift or energy. Neutral stick means approximately the augmented trim/1-G demand for the current state, not a flight-path-attitude hold. All commands still pass through mixer, hydraulics, actuators, and aerodynamics.

8.4 AUG_OFF (DIRECT) mode

Pitch → elevator/stabilator demand. Roll → lateral-surface demand. Yaw → rudder demand. Trim remains. Hydraulics, actuator rates, stops, damage, and actual surface response remain. Control mixer/gearing remains. No closed-loop G-command, no roll-rate-command loop, no AoA/G limiter, no artificial FCS stability damping. Natural static and dynamic aerodynamic stability supply the airplane's behaviour.

8.5 ADLC overlay and priority with AUG_OFF

ADLC is an approach-control overlay. When engaged it may command on-speed AoA through its longitudinal/throttle path and allow flight-path-angle input; lateral control remains that of the underlying FCS mode/status. Selecting AUG_OFF while ADLC is engaged first disengages ADLC, then transfers to DIRECT. ADLC cannot remain as a hidden closed-loop longitudinal controller under AUG_OFF. If required augmentation capability is lost, ADLC fails or disengages. Disengagement returns to the previously active underlying mode/status with no snap of attitude, surface position, trim, G, or velocity.

8.6 Bumpless FCS mode transfer

Mode or capability changes must not produce discontinuous changes to actual surfaces, commanded surfaces, attitude, angular rate, G, or velocity solely because the mode or status changed.

AUGMENTED → AUG_OFF: preserve actual surface positions. Introduce a bounded temporary transfer bias equal to the difference between the last augmented surface command and the direct stick-plus-trim command at the instant of transfer. The temporary transfer bias is a distinct state variable, separate from PilotTrimState. It shall not modify, accumulate into, or silently redefine pilot trim. It decays deterministically to zero over a short interval while the existing pilot trim remains unchanged. Freeze or safely reset augmentation integrators. No instantaneous surface movement.

AUG_OFF → AUGMENTED: initialise feedback and integrator state from current aircraft state; capture current trim condition; engage protection continuously; do not command an instantaneous recovery manoeuvre.

8.7 Autothrottle

Conventional autothrottle is independent of FCS mode/status and of ADLC. It captures KIAS and modulates through military power. ADLC has its own autothrottle behaviour when engaged. Afterburner overrides until the pilot recaptures speed.

9. Variable-Geometry Wing Sweep

State: sweep_cmd, sweep_left_actual, sweep_right_actual. Automatic Mach/flight-condition schedule commands both sides identically; manual override and return-to-Auto exist. Hydraulics rate-limit actual positions. Asymmetry from damage is consumed by aerodynamics. Coefficient tables and the inertia schedule are indexed by sweep.

10. Actuators, Hydraulics, Mass Properties, Ground Effect

All surface and sweep commands pass through mixer, hydraulic authority, and rate limits. Mass changes with fuel, missiles, and ammunition; inertia from the sweep×mass schedule under the Appendix A.2 tensor convention. Ground effect is required: a deterministic table versus height/span modifies induced drag and may modify lift. Exact table values are Phase-2 gated.

11. Integration, Clamps, and Tick Order

Semi-implicit Euler is the baseline candidate; compare with RK2 on hard trajectories. Choose the cheapest integrator inside the approved trajectory-error envelope (tolerances Phase-2 gated). Clamps are overflow/fault guards. Tick order: actual surfaces → atmosphere sample and forces → aircraft integration (body rates, body velocity, body→NED, position) → weapon integration.

12. Aircraft and Weapon Models

Player and combat AI share the identical 6DOF pipeline and tables. Guided missiles are 100 Hz 3DOF point-mass with mass, thrust/burn, gravity, atmosphere, drag, lift/turn demand, guidance, and mandatory maneuver-energy loss; long-range uses deterministic energy-seeking loft. SAMs share the core. Ballistic weapons omit powered guidance after boost. Cannon uses pooled groups. All sample the shared atmosphere and wind service.

13. Fixed-Point Strategy and Dual Oracle

Numeric registry is sole authority. Dual reference: high-precision physics oracle (tolerance) and bit-exact fixed-point host reference (exact) → production C → optional measured ASM.

14. Cycle Budget and Deterministic Table Residency

Nine aircraft + sixteen guided missiles on the 100 Hz path. Hot tables for live types are resident under a deterministic mission-load plan before the sortie or before the type becomes live.

15. Validation and AUG-OFF Flyability

Golden vectors cover atmosphere/wind, airspeed conversions, static and dynamic coefficients, sweep, closed-loop integration, engine force/moment, AUGMENTED / AUG_OFF under NORMAL / DEGRADED / DIRECT_ONLY, bumpless transfer (including transfer-bias decay without trim mutation), stall/separation state, ground effect, mass change, and missile loft. Phase-2 artifact F65_UNAUGMENTED_FLIGHT_ENVELOPE defines the bounds within which AUG_OFF controllability is required. AUG-OFF FLYABILITY: a healthy F-65A with ADLC OFF and FCSMode = AUG_OFF (capability

permitting) remains manually controllable without hidden G/rate command, altitude hold, artificial damping, or non-physical authority throughout that envelope.

16. Designer and Engineer Roles

Designers author coefficient tables, schedules, F65_UNAUGMENTED_FLIGHT_ENVELOPE, and missile tables (Phase-2/3 gates). Engineers implement atmosphere/wind, aero evaluation, closed-loop 6DOF integrator, engine force/moment, FCSMode/FCSStatus manager, control mixer, bumpless transfer with distinct transfer-bias state, ADLC overlay, autothrottle, actuators/hydraulics, and 3DOF missiles. C first; measured ASM only.

17. Worked Example — Fictitious Aircraft (6DOF, Imperial)

Weight $W = 40\,000$ lbf $\rightarrow$ mass $m = W/g \approx 1\,243$ slug. $S = 480$ ft², $c_{\text{m}} = 12$ ft, $b = 40$ ft. Sweep both sides = 25° . Inertia from 25° /normal-mass bin under the Appendix A.2 tensor convention. Actual $\delta e = -2^\circ$, $\delta a = +1.5^\circ$, $\delta r = 0^\circ$. Engines offset ± 3 ft. $h = 26\,000$ ft, TAS after wind ≈ 440 kt, Mach ≈ 0.747 , $q_{\text{m}} \approx 313$ psf, $\alpha \approx -4.8^\circ$, $\beta \approx 2.1^\circ$. Tables plus rate damping and engine installation moments feed the complete body-axis equations; candidate integrator advances rates, body velocity, body $\rightarrow$ NED, and position. Illustrative only.

18. Worked Example — Guided Missile (3DOF, Imperial)

Launch weight 400 lbf, burnout 240 lbf, burn 6 s, boost 4 000 lbf. At $t = 3.5$ s, weight 320 lbf, $h = 31\,000$ ft, TAS after wind ≈ 980 kt, Mach ≈ 1.55 , $q_{\text{m}} \approx 1\,350$ psf. Guidance 12 g (unsaturated); induced drag of turning lift added. Integration on the shared 100 Hz path.

Appendix A — Complete Reference Equations (Closed Kinematic Loop)

Continuous forms for the high-precision oracle. Production is discrete fixed-point at 100 Hz.

A.1 Translation (body axes)

$$\begin{aligned} \dot{u} &= r v - q w + F_x / m \\ \dot{v} &= p w - r u + F_y / m \\ \dot{w} &= q u - p v + F_z / m \end{aligned}$$

A.2 Rotation — inertia tensor convention and equations

The product-of-inertia sign convention used by this paper is:

$$I = \begin{bmatrix} I_{xx} & 0 & -I_{xz} \\ 0 & I_{yy} & 0 \\ -I_{xz} & 0 & I_{zz} \end{bmatrix}$$

Every authored or interpolated mass-property state must satisfy the positive-definite constraints:

$$I_{xx} > 0, I_{yy} > 0, I_{zz} > 0, I_{xx}I_{zz} - I_{xz}^2 > 0.$$

An inertia table that violates these constraints is rejected by the host-oracle / mission-load path and must not reach the fixed-point solve.

With quasi-static I (dI/dt omitted on target) the moment equations consistent with the tensor above are:

$$\begin{aligned} L &= I_{xx} p - I_{xz} r - I_{xz} p q - (I_{yy} - I_{zz}) q r \\ M &= I_{yy} q + (I_{xx} - I_{zz}) p r + I_{xz} (p^2 - r^2) \\ N &= I_{zz} r - I_{xz} p + I_{xz} q r - (I_{xx} - I_{yy}) p q \end{aligned}$$

Solve the linear system for p, q, r each tick (bounded fixed-point solve). When $I_{xz} = 0$ the equations reduce to the diagonal form. The high-precision oracle may retain full dI/dt terms for validation of the quasi-static approximation.

A.3 Atmosphere, wind, aero

$$\begin{aligned} V_{\text{air}} &= V_g - W(h) \\ q &= \frac{1}{2} \rho |V_{\text{air}}|^2 \\ C &= C_{\text{base}}(\alpha, \beta, \text{Mach}, \text{sweep}, \text{config}) + C_{\delta} + C_{\text{rate}}(p, q, r) + C_{\text{damage}} + C_{\text{asym}} + \text{ground-effect terms} \end{aligned}$$

A.4 Engine force and moment

$$\begin{aligned} F &= \sum F_i \\ M &= \sum (r_i \times F_i) \end{aligned}$$

A.5 Quaternion rate

$$\dot{q} = \frac{1}{2} \Omega(\omega) q, \text{ periodic normalisation with widened intermediates.}$$

A.6 Body → NED velocity and position integration

$$\begin{aligned} V_{\text{NED}} &= R_{\text{body} \rightarrow \text{NED}} \cdot V_{\text{body}} = (V_N, V_E, V_D) \\ \dot{h} &= V_N, \dot{h} = V_E, \dot{h} = -V_D \text{ (altitude positive up; } V_D \text{ positive down)} \end{aligned}$$

Sector crossing is handled by the world-position service. The complete authoritative chain each tick is: forces/moments → body accelerations → body rates and body velocity → attitude update → body→NED transform → NED/local position update.

Appendix B — References and Configuration-Control Notes

B.1 Project authority

- F-65 Megawing Revision 1.5.1 — Architecture Invariants
- F-65 Megawing Gameplay and Simulation Requirements Supplement Draft 0.2
- F-65 Engine Runtime and Toolchain Design Supplement Draft 0.2
- F-65 Technical Alignment and Read-First Supplement v1.0
- MEGA65 Flight Simulation 3D Graphics White Paper

B.2 Configuration-control posture for FCS terminology

This frozen paper defines $\text{FCSMode} = \text{AUGMENTED} / \text{AUG_OFF}$ (commanded/pilot-selected) and $\text{FCSStatus} = \text{NORMAL} / \text{DEGRADED} / \text{DIRECT_ONLY}$ (capability) as the engineering contract. Architecture and Gameplay must be revised under configuration control to replace Assisted/Manual rate-command language with these records, to introduce the non-augmenting control mixer, bumpless transfer with a distinct decaying transfer-bias state that does not mutate PilotTrimState , ADLC-as-overlay with AUG_OFF priority, and $\text{F65_UNAUGMENTED_FLIGHT_ENVELOPE}$. Phase 1 interface generation follows this paper's shapes. Older Assisted/Manual field names are not to be frozen as the public Phase-1/Phase-2 contract.

B.3 Technical sources

- Etkin & Reid; Stevens, Lewis & Johnson; Nelson — rigid-body dynamics, static and dynamic stability
- JSBSim / FlightGear table organisation; International Standard Atmosphere
- Proportional navigation and energy-seeking loft literature; air-breathing engine performance modelling
- Variable-geometry wing aerodynamics

B.4 Vintage references

- Chuck Yeager's Advanced Flight Trainer
- Artwick / early SubLOGIC-MSFS lineage
- A-10 Cuba!, F/A-18 Korea, Falcon 3.0

B.5 Platform

- MEGA65 Chipset Reference and 45GS02 programming practice

End of Engineering White Paper v3.3 — FROZEN governing model for Phase-2 F-65 6DOF physics: flyable physical airframe under AUGMENTED and AUG_OFF control, control mixer, bumpless transfer with distinct transfer-bias state, atmosphere, engines, sweep, closed kinematic loop, and explicit inertia-tensor convention on the MEGA65.